

Module 9: Inheritance Genetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. An organism's genotype is its:
2. A. set of observable physical traits
3. B. surrounding environment
4. C. combination of alleles
5. D. total number of cells
6. Answer: C

7. Why: Genotype is the combination of alleles an organism carries. Phenotype is the observable trait that results.

8. Two heterozygotes are crossed ($Aa \times Aa$). What fraction of offspring are expected to show the recessive trait?
9. A. $1/4$
10. B. $1/2$
11. C. $3/4$
12. D. 0
13. Answer: A

14. Why: $Aa \times Aa$ gives a 1 AA : 2 Aa : 1 aa ratio. Only the aa offspring ($1/4$) show the recessive trait.

15. A heterozygote (Aa) shows the dominant trait because:
16. A. it carries no recessive allele
17. B. recessive alleles are destroyed
18. C. both alleles are always expressed equally
19. D. one dominant allele is enough to mask the recessive allele
20. Answer: D

21. Why: In simple dominance, one dominant allele determines the phenotype. The recessive allele is present but masked.
22. What is a pedigree used for?
23. A. measuring the size of a cell
24. B. tracing how a trait passes through a family to infer its inheritance
25. C. editing genes directly
26. D. counting chromosomes in soil samples
27. Answer: B
28. Why: A pedigree diagrams family relationships and phenotypes so you can reason about how a trait is inherited.